GE MEDICAL SYSTEMS

Technical Publications

Direction 46-305150
Revision 1

Signa®
0.5T Anterior Neck Surface Coil

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, nature, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

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Complete instructions regarding claim procedures are found in section “S” of the Policy & Procedure Bulletins.

3/12/92

GE MEDICAL SYSTEMS

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LIST OF EFFECTIVE PAGES

REVISION	DATE	PRIMARY REASON FOR CHANGE
REV 0	5/18/93	Initial Release
REV 1	8/2/93	Corrected 46- #, Corrected Coupling,Decoupling images

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1-0 INTRODUCTION

The proper positioning and use of the Anterior Neck Surface Coil is important to ensure quality images and the safety of the patient. This is a receive only coil designed for proton imaging on GE SIGNA® ADVANTAGE™ 0.5T Systems only. This coil is recommended for Anterior Neck studies and has been tuned (electronically matched) for that anatomical region.

2-0 SIGNA® ADVANTAGE™ 0.5T SCANNER VERIFICATION

2-1 Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling.

NOTE

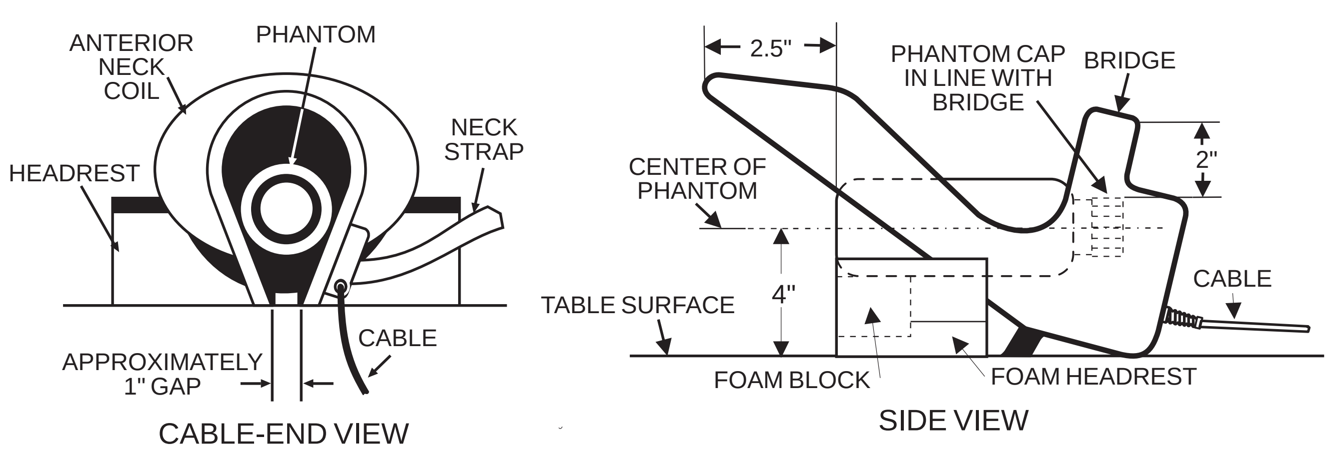
The surface coil decoupling phantom must be filled with distilled water and the supplied cupric sulfate prior to use for any imaging tests. To properly prepare the phantom, completely fill the one liter Nalgene bottle supplied with the coil with distilled water while mixing in the vial of cupric sulfate. Ensure that all of the cupric sulfate has dissolved, and that there are no air bubbles when the cap is replaced on the bottle.

2-2 Select [**NEW EXAM**] to enable the patient table **LANDMARK** function. Locate the surface coil decoupling phantom on the patient table and position as shown in ILLUSTRATION 2-1. Do not place the Anterior Neck Coil under the decoupling phantom. Raise the decoupling phantom to the same approximate location as the surface coil. **LANDMARK** on the center of the decoupling phantom.

2-3 Set up the scan on the SIGNA ADVANTAGE 0.5T system using the parameters in ILLUSTRATION 2-2. Select **AUTO PRESCAN** to calibrate the RF power level. Record the value for TG for future use in Section 3.

2-4 Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.

2-5 For the SNR comparisons, select [**CANCEL**], [**NEW SERIES**], and acquire two identical images using the scan parameters in ILLUSTRATION 2-3. After acquiring the first image, select**SCAN** to obtain a second. Note the series number for future reference.



DECOUPLING PHANTOM PLACEMENT

ILLUSTRATION 2-1

PATIENT EXAM INFORMATION

ID:	**GESERVICE**
Name:	
Patient Weight:	**111**
	[**Patient Position**]

PATIENT POSITION

Patient Entry:	[**Head First**]
Patient Position:	[**Supine**]
Axial/Sag Landmark:	[**Sternal Notch**]
Coil Type:	[**Body Coil**]
Scan Plane:	[**Axial**]
	[**Imaging Parameters**]

IMAGING PARAMETERS

Image mode:	[**2D**]
(SAR must be **ON** *)*	[**Monitor SAR**]
Pulse Sequence:	[**Spin Echo]**
Imaging Options:	[**None**]
or enter PSD Filename:	
	[**Next Screen**] or [**Scan Timing**]

SCAN TIMING

Number of Echos:	[**4**]
Echo Time (TE):	[**20 ms**]
Rep Time (TR):	[**OTHER**] **600 ms**
	[**Scan Set—Up**] *(Release 5.3)*
	[**Scanning Range**] *(Release 5.2)*

SCAN SET—UP *(Release 5.3)*

Auto CF:	[**Water**]
Receive Bandwidth:	*for 0.5T only* [**16kHz**]
	[**Scanning Range**]

SCANNING RANGE

Field of View:	[**32**]
Slice Thickness:	[**5 mm**]
Interscan Spacing:	[**0**]
Start Location (I/S):	**0**
End Location (I/S):	**0**
No of Scan Locations	**1**
FOV Center (L/R)	**0** (P/A): **0**
	[**<**] [**Acquisition Time**]

ACQUISITION TIME

Acq. Matrix: (freq.)	[**256**]
Acq. Matrix: (phase)	[**128**]
Frequency Direction:	[**R/L**]
Phase FOV:	default *(Release 5.3)*
Imaging Time:	[**1 Nex**]
Contrast:	[**NO**]
Table Delta	**0**
	[**Scan Ops**] *(Release 5.3)*
	[**Scan Set—Up**] *(Release 5.2)*

SCAN SET—UP

Auto CF:	[**Water**]
Receive B/W (+/-):	*for 0.5T ONLY* [**16 KHz**]
	[**Scan Ops**]

SCAN OPERATIONS

SYSTEM TEST SCAN PARAMETERS

ILLUSTRATION 2-2

PATIENT EXAM INFORMATION

ID:	**GESERVICE**
Name:	
Patient Weight:	**111**
	[**Patient Position**]

PATIENT POSITION

Patient Entry:	[**Feet First**]
Patient Position:	[**Supine**]
Axial/Sag Landmark:	[**Sternal Notch**]
Coil Type:	[**Body Coil**]
Scan Plane:	[**Axial**]
	[**Imaging Parameters**]

IMAGING PARAMETERS

Image mode:	[**2D**]
(SAR must be **ON** *)*	[**Monitor SAR**]
Pulse Sequence:	[**Spin Echo]**
Imaging Options:	[**None**]
or enter PSD Filename:	
	[**Next Screen**] or [**Scan Timing**]

SCAN TIMING

Number of Echos:	[**1**]
Echo Time (TE):	[**20 ms**]
Rep Time (TR):	[**OTHER**] **600 ms**
	[**Scan Set—Up**] *(Release 5.3)*
	[**Scanning Range**] *(Release 5.2)*

SCAN SET—UP *(Release 5.3)*

Auto CF:	[**Water**]
Receive Bandwidth:	*for 0.5T only* [**16kHz**]
	[**Scanning Range**]

SCANNING RANGE

Field of View:	[**32**]
Slice Thickness:	[**5 mm**]
Interscan Spacing:	[**0**]
Start Location (I/S):	**0**
End Location (I/S):	**0**
No of Scan Locations	**1**
FOV Center (L/R)	**0** (P/A): **0**
	[**<**] [**Acquisition Time**]

ACQUISITION TIME

Acq. Matrix: (freq.)	[**256**]
Acq. Matrix: (phase)	[**128**]
Frequency Direction:	[**R/L**]
Phase FOV:	default *(Release 5.3)*
Imaging Time:	[**1 Nex**]
Contrast:	[**NO**]
Table Delta	**0**
	[**Scan Ops**] *(Release 5.3)*
	[**Scan Set—Up**] *(Release 5.2)*

SCAN SET—UP

Auto CF:	[**Water**]
Receive B/W (+/-):	*for 0.5T ONLY* [**16 KHz**]
	[**Scan Ops**]

SCAN OPERATIONS

BODY COIL SNR SCAN PARAMETERS

ILLUSTRATION 2-3

3-0 ANTERIOR NECK COIL IMAGING PERFORMANCE VERIFICATION

3-1 Set up the SIGNA ADVANTAGE 0.5T Anterior Neck coil and decoupling phantom on the patient table as shown in ILLUSTRATION 2-1. Connect the surface coil cable to the **SURFACE COIL QUICK DISCONNECT BOX** and connect to the **HEAD COIL PORT**.

3-2 Select [**CANCEL**], [**NEW SERIES**], and repeat the scan using the parameters in ILLUSTRATION 3-1. Use **AUTO PRESCAN** to set the values of **R1** and **R2**. If the value of TG does not match the value obtained in step 2-3, use **MANUAL PRESCAN** to set the value of TG to the value found in step 2-3.

3-3 Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images. Study the third and fourth echoes using a **WINDOW WIDTH** and **WINDOW LEVEL** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-2. A decoupling failure will cause holes, distortions, or striped patterns in the image as seen in ILLUSTRATION 3-3. Refer to Section 4-0 to troubleshoot decoupling failures.

3-4 Select [**CANCEL**], [**NEW SERIES**], and acquire two identical images using the scan parameters in ILLUSTRATION 3-4. After acquiring the first image, select **SCAN** to obtain a second.

3-5 Compare the signal to noise ratio of the Anterior Neck Coil image of step 3-4 with the signal to noise ratio of the body coil image of step 2-5. See Section 4-0, *SNR Image Analysis*. The signal to noise ratio of the Anterior Neck Coil should be 2.5 times greater than the body coil signal to noise ratio.

3-6 Inspect the images for visible ghosting [similar in appearance to motion artifacts] in the phase encoding direction. If images contain ghosting artifacts, an intermittent cable or connection is suspected. Refer to Section 5-0 for troubleshooting information.

PATIENT EXAM INFORMATION
ID: **GESERVICE**
Name:
Patient Weight: **111**
[**Patient Position**]

PATIENT POSITION
Patient Entry: [**Head First**]
Patient Position: [**Supine**]
Axial/Sag Landmark: [**Sternal Notch**]
Coil Type: [**Other**] [**Anterior Neck**]
Scan Plane: [**Sagittal**]
[**Imaging Parameters**]

IMAGING PARAMETERS
Image mode: [**2D**]
(SAR must be **ON** *)* [**Monitor SAR**]
Pulse Sequence: [**Spin Echo]**
Imaging Options: [**None**]
or enter PSD Filename:
[**Next Screen**] or [**Scan Timing**]

SCAN TIMING
Number of Echos: [**4**]
Echo Time (TE): [**20 ms**]
Rep Time (TR): [**OTHER**] **600 ms**
[**Scan Set—Up**] *(Release 5.3)*
[**Scanning Range**] *(Release 5.2)*

SCAN SET—UP *(Release 5.3)*
Auto CF: [**Water**]
Receive Bandwidth: *for 0.5T only* [**16kHz**]
[**Scanning Range**]

SCANNING RANGE
Field of View: [**32**]
Slice Thickness: [**5 mm**]
Interscan Spacing: [**0**]
Start Location (I/S): **0**
End Location (I/S): **0**
No of Scan Locations **1**
FOV Center (L/R) **0** (P/A): **0**
[**<**] [**Acquisition Time**]

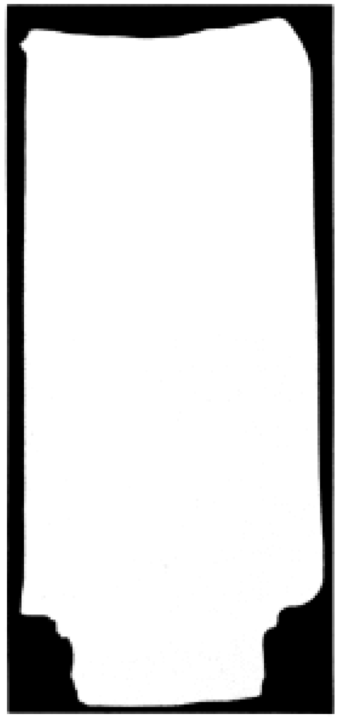
ACQUISITION TIME
Acq. Matrix: (freq.) [**256**]
Acq. Matrix: (phase) [**128**]
Frequency Direction: [**R/L**]
Phase FOV: default *(Release 5.3)*
Imaging Time: [**1 Nex**]
Contrast: [**NO**]
Table Delta **0**
[**Scan Ops**] *(Release 5.3)*
[**Scan Set—Up**] *(Release 5.2)*

SCAN SET—UP
Auto CF: [**Water**]
Receive B/W (+/-): *for 0.5T ONLY* [**16 KHz**]
[**Scan Ops**]

SCAN OPERATIONS

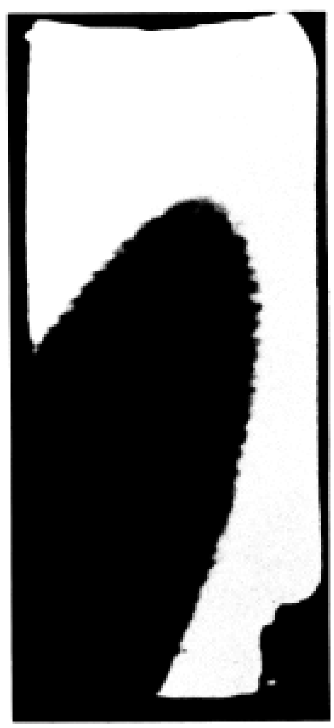
SURFACE COIL DECOUPLING TEST SCAN PARAMETERS

ILLUSTRATION 3-1



NORMAL DECOUPLING

ILLUSTRATION 3-2



DECOUPLING FAILURE

ILLUSTRATION 3-3

PATIENT EXAM INFORMATION

ID:	**GESERVICE**
Name:	
Patient Weight:	**111**
	[**Patient Position**]

PATIENT POSITION

Patient Entry:	[**Head First**]
Patient Position:	[**Supine**]
Axial/Sag Landmark:	[**Sternal Notch**]
Coil Type:	[**Other**] [**Anterior Neck**]
Scan Plane:	[**Axial**]
	[**Imaging Parameters**]

IMAGING PARAMETERS

Image mode:	[**2D**]
(SAR must be **ON** *)*	[**Monitor SAR**]
Pulse Sequence:	[**Spin Echo]**
Imaging Options:	[**None**]
or enter PSD Filename:	
	[**Next Screen**] or [**Scan Timing**]

SCAN TIMING

Number of Echos:	[**1**]
Echo Time (TE):	[**20 ms**]
Rep Time (TR):	[**OTHER**] **600 ms**
	[**Scan Set—Up**] *(Release 5.3)*
	[**Scanning Range**] *(Release 5.2)*

SCAN SET—UP *(Release 5.3)*

Auto CF:	[**Water**]
Receive Bandwidth:	*for 0.5T only* [**16kHz**]
	[**Scanning Range**]

SCANNING RANGE

Field of View:	[**32**]
Slice Thickness:	[**5 mm**]
Interscan Spacing:	[**0**]
Start Location (I/S):	**0**
End Location (I/S):	**0**
No of Scan Locations	**1**
FOV Center (L/R)	**0** (P/A): **0**
	[**<**] [**Acquisition Time**]

ACQUISITION TIME

Acq. Matrix: (freq.)	[**256**]
Acq. Matrix: (phase)	[**128**]
Frequency Direction:	[**R/L**
Phase FOV:	default *(Release 5.3)*
Imaging Time:	[**1 Nex**]
Contrast:	[**NO**]
Table Delta	**0**
	[**Scan Ops**] *(Release 5.3)*
	[**Scan Set—Up**] *(Release 5.2)*

SCAN SET—UP

Auto CF:	[**Water**]
Receive B/W (+/-):	*for 0.5T ONLY* [**16 KHz**]
	[**Scan Ops**]

SCAN OPERATIONS

ANTERIOR NECK COIL SNR SCAN PARAMETERS
ILLUSTRATION 3-4

4-0 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

4-1 Procedure

4-1-1 Touch [**UTILITIES**], [**MRTools**], [**Image Quality**], then [**SNRTest**]. The SNR Test screen is displayed. See ILLUSTRATION 4-1.

4-1-2 Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select [**List Exams**], [**List Series**], or [**List Images**] to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

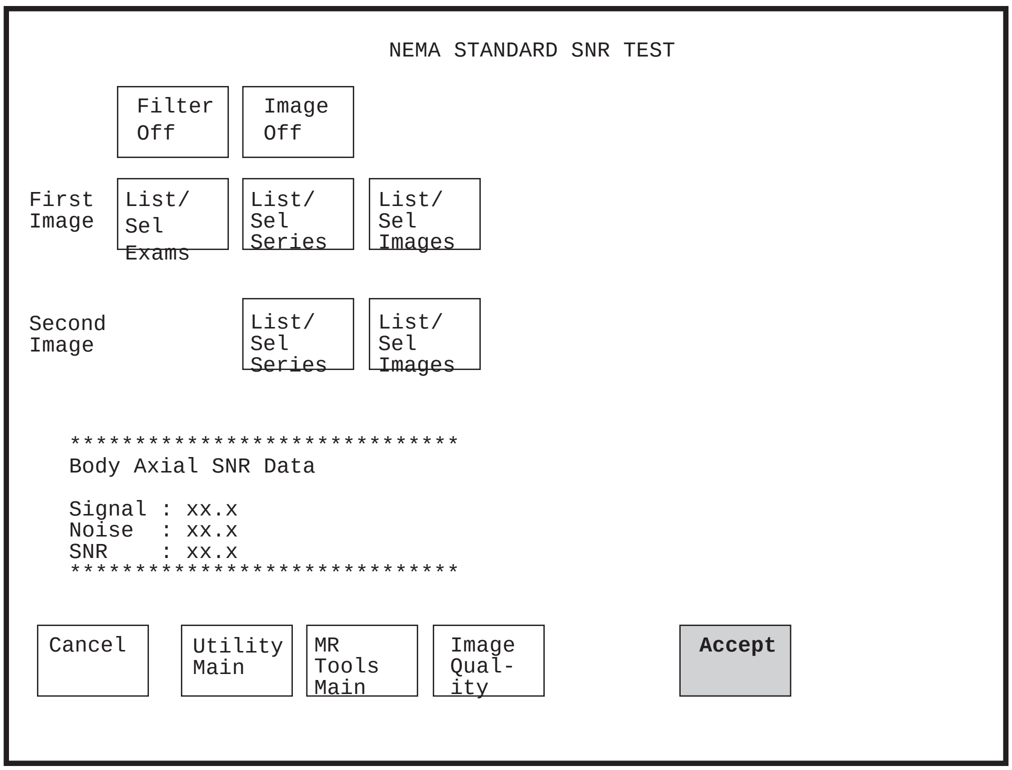
4-1-3 Enter second image series and image numbers. If series or image numbers are not known, select [**List Series**] or [**List Images**] to display list. The second image Exam Number is defaulted to the same as first image.

4-1-4 Select [**Filter Off**] and [**Image Off**].

4-1-5 Touch [**Accept**] to begin analysis. The final values are displayed on the screen. See ILLUSTRATION 4-1.

4-1-6 Record Signal to Noise Ratio on ILLUSTRATION 4-2.

4-1-7 Touch [**Continue**] then select the next exam and repeat the above analysis for each image pair.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 4-1

Coil	Signal to Noise Ratio	Date
Body Coil		
Anterior Neck Coil		

SNR VALUES
ILLUSTRATION 4-2

5-0 TEST AND TROUBLESHOOTING PROCEDURES

5-1 Coil Impedance Measurements

5-1-1 Move to an area where the magnetic field from the SIGNA ADVANTAGE 0.5T magnet is less than 5 Gauss. Load the surface coil with a human subject. Use the procedure in the OPERATION MANUAL to place the surface coil.

5-1-2 Connect the surface coil cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 21.29 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.

5-1-3 The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 40 to 60 ohms under test conditions.

5-1-4 The phase angle should nominally be zero degrees. It should fall within the range of -20 to +20 degrees under test conditions.

5-2 Coil Integrity Tests

5-2-1 Repeat steps 5-1-1 to 5-1-4 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Using the **HIGH SPEED** mode on the vector impedance meter, note any erratic fluctuations in the coil impedance . Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 6-0 for service procedures.

5-3 Decoupling Diode Tests

5-3-1 Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.4 and 0.9. Reversing the leads should cause the reading to be infinity [overrange].

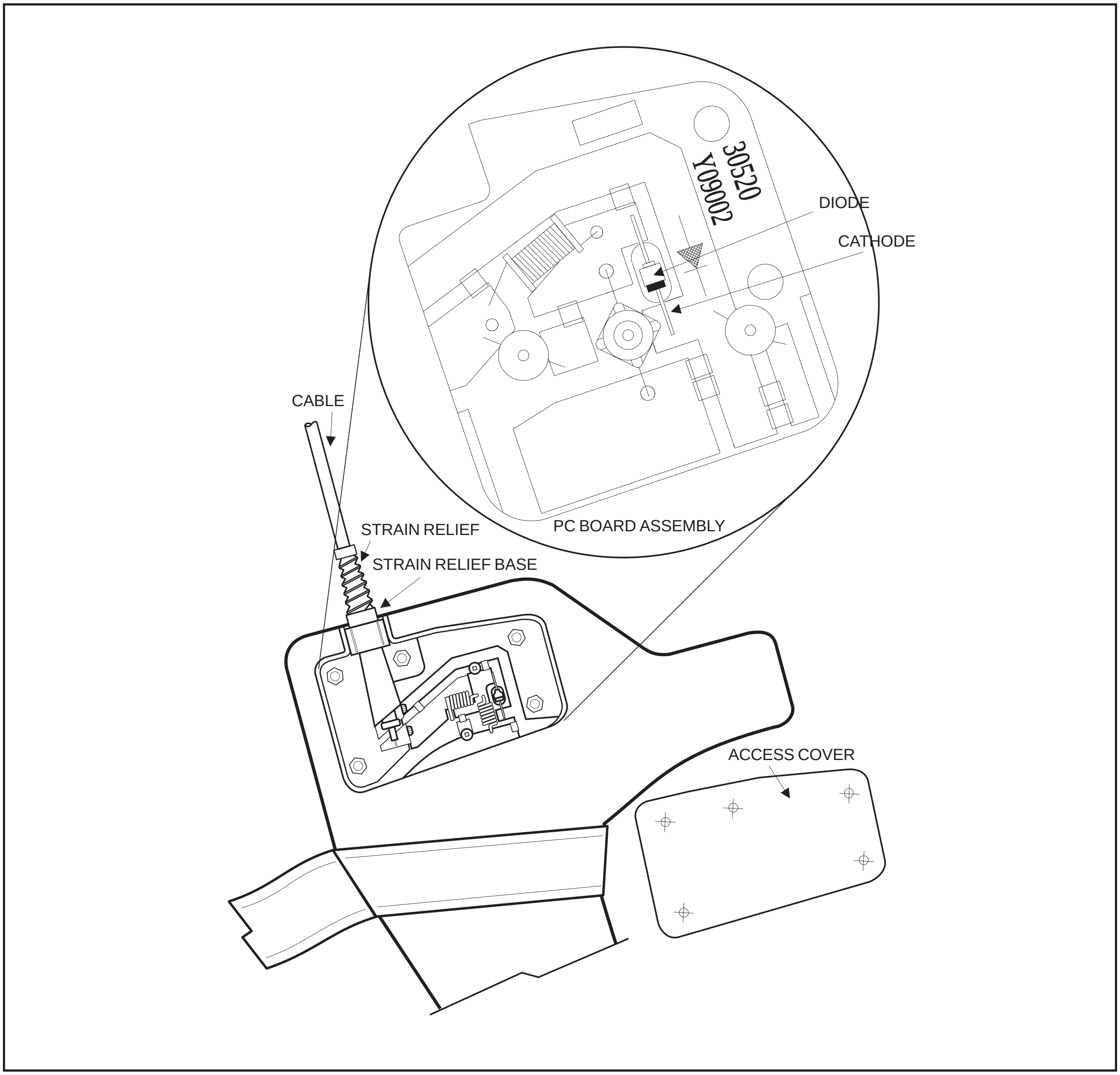
5-3-2 An open reading in both directions indicates an open diode, or an open cable. Refer to Section 5-4 to check the cable. Refer to Section 6-4 to replace the diode.

5-3-3 A short circuit or excessively low reading indicates a shorted diode or a shorted cable. Refer to Section 5-4 to check the cable. Refer to Section 6-4 to replace the diode.

5-4 Cable Tests

5-4-1 If the surface coil passes the tests in Sections 5-1, 5-2, and 5-3, the cable can be assumed to be functional.

5-4-2 If a short or open circuit was discovered in Section 5-3, the cable may be tested by disconnecting the cable at the PC board via the service access cover, as shown in ILLUSTRATION 5-1. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continuity test function.



ACCESS COVER, INTERNAL CABLE CONNECTION

ILLUSTRATION 5-1

6-0 SERVICE PROCEDURES

6-1 A service kit is available. ILLUSTRATION 6-1 describes the contents of the kit. No other components are replaceable.

CABLE KIT 46-328440P1 — Consisting of the following items:

ITEM	DESCRIPTION	QTY
1	Cable Assembly	1
2	Screws	5

1	Anterior Neck Coil with Cable—46-328282P1	1

SERVICE KIT

ILLUSTRATION 6-1

6-2 Coaxial Cable Replacement - ILLUSTRATION 6-2

6-2-1 Remove the service access cover from the housing by removing the nylon screws attaching the cover.

6-2-2 Unscrew the three stand-offs from the cable/PCB assembly.

6-2-3 Scrape off the conformal coating from two soldered connection tabs which join the cable/PCB assembly to the decoupling board.

6-2-4 Unsolder the connection tabs and remove cable/PCB assembly.

6-2-5 Place the new cable/PCB assembly on the screwposts. Do not tighten the offsets.

6-2-6 Solder the tabs, then tighten the offsets.

6-2-7 Attach the service access cover to the coil housing.

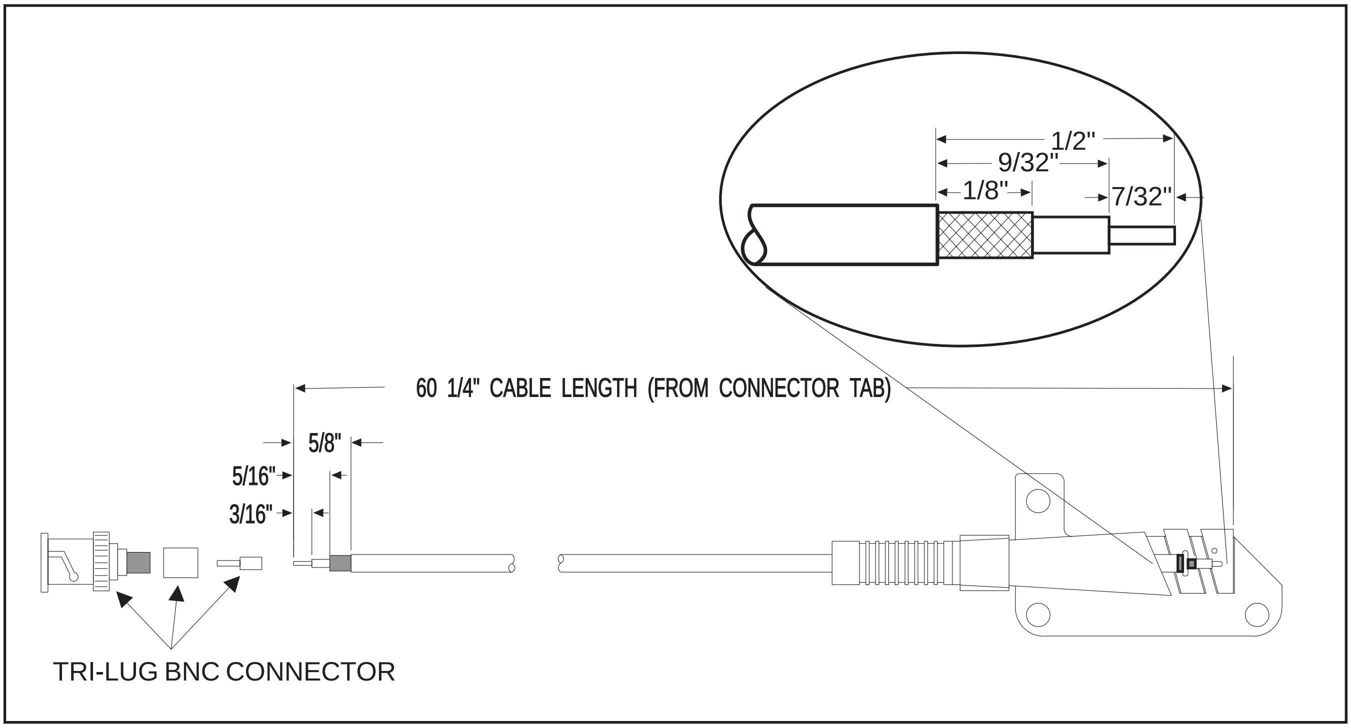
6-2-8 Repeat the tests in Sections 3-0 and 4-0 to verify proper coil operation.

6-3 Output Connector Replacement - ILLUSTRATION 6-2

6-3-1 If the cable will be shortened to less than 60-1/8 inches by a repair, it is recommended that the entire cable be replaced in the event of a connector or connector-end cable failure.

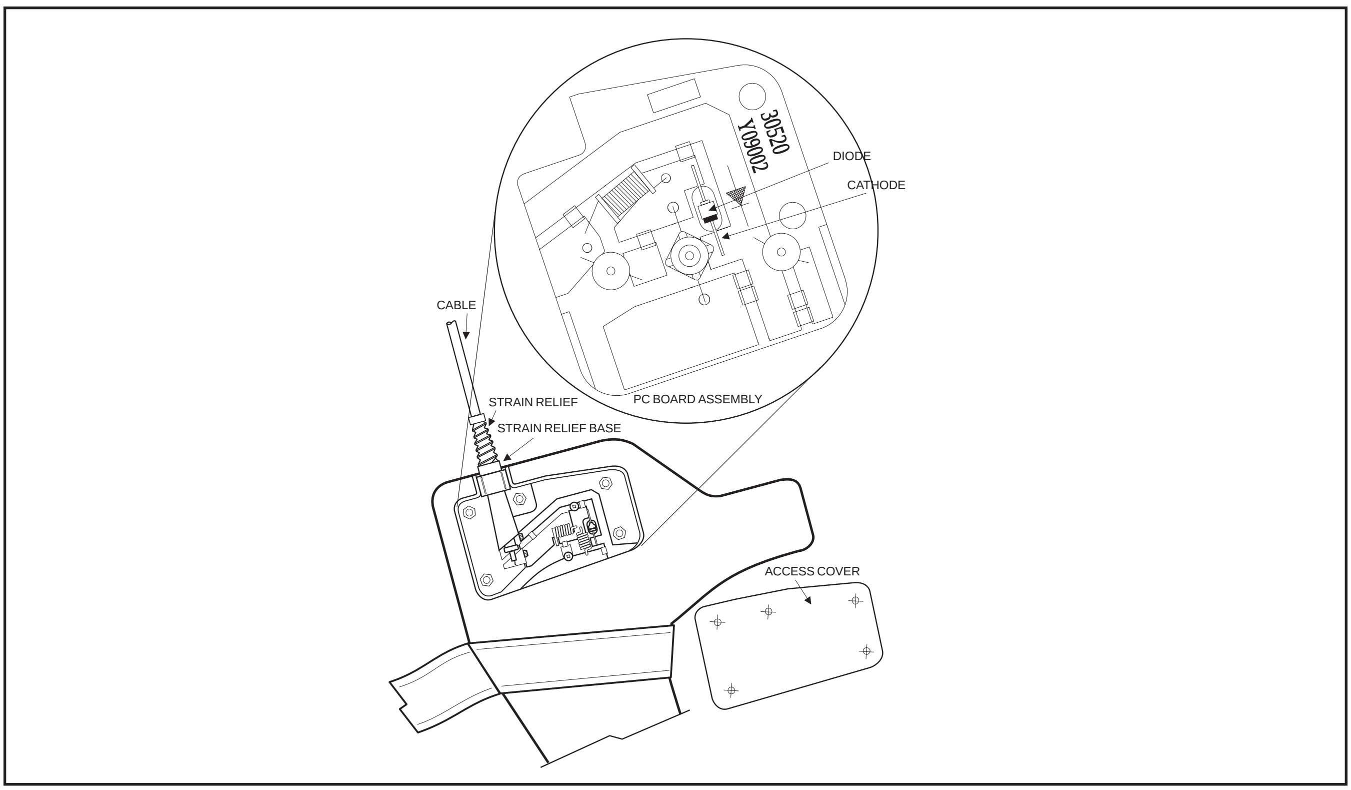
6-3-2 Replace the connector using ILLUSTRATION 6-2 as a guide. Use a new Tri-Lug BNC coaxial fitting - part number 46-317498P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.

6-3-3 Repeat the tests in Sections 3-0 and 4-0 to verify proper coil operation.



COAXIAL CABLE REPLACEMENT

ILLUSTRATION 6-2



DIODE LOCATION

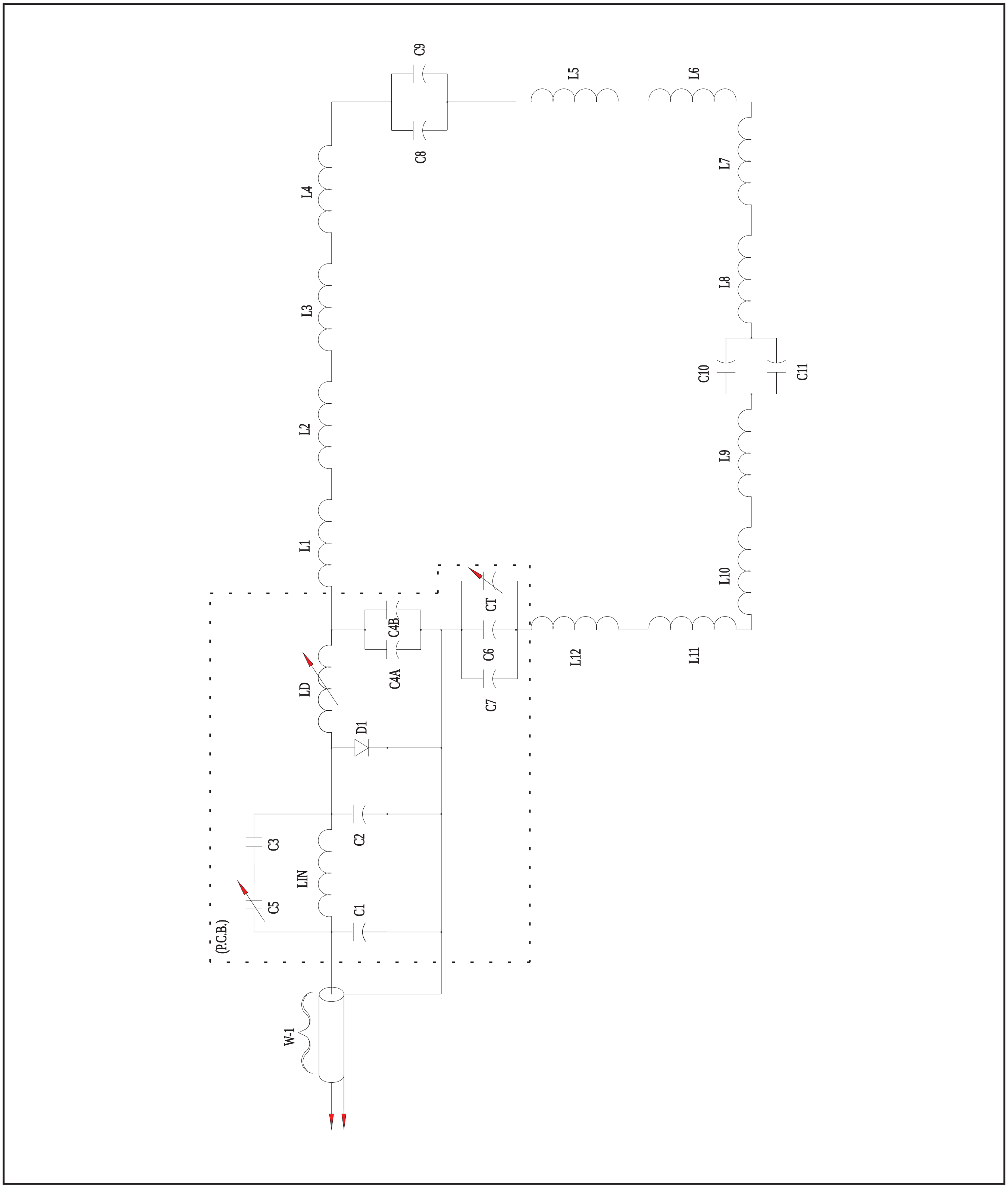
ILLUSTRATION 6-3

6-4 Diode Replacement

6-4-1 If it has been determined that a diode must be replaced - Section 5-3, remove the access cover.

6-4-2 Diode placement is shown in ILLUSTRATION 6-3 with the schematic shown as ILLUSTRATION 6-4.

6-4-3 Unsolder defective diode and replace with new diode - part number 46-221735P1.



SCHEMATIC OF ANTERIOR NECK COIL

ILLUSTRATION 6-4

SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 4, *SNR IMAGE ANALYSIS.* After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2, enter the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: ______________________

Anterior Neck Coil Value: ____________________

Date: ____________

SNR DATA QA (QUALITY ASSURANCE) CHECK:

1 Date	2 Anterior Neck QA SNR Value:	3 Is new value divided by original value > 85%	
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N
			Y/N

Make additional copies of this document as needed.

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

For proper assessment of defective returned coils, both sides of this form should be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: ____________________

SITE NAME: ____________________

SITE ADDRESS: ____________________

____

SERVICE ENGINEER: ____________________

COIL SERIAL NUMBER: ____________________

DATE COIL INSTALLED: ____________________

DESCRIPTION OF COIL PROBLEM: ____________________
[Please be descriptive; “Broken” is not enough] ____________________

____

ELECTRICAL CHECKS VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE—Refer to Section 5-1.	
MAGNITUDE	
PHASE	
VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE—Refer to Section 5-1.	
MAGNITUDE	
PHASE	
PIN DIODE TEST—Refer to Section 5-3.	
DIODE DROP—FORWARD BIAS	
DIODE DROP—REVERSE BIAS	

TLT Procedure Table

Note

An alternate proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" calibration procedure located in Direction 15065, *Signa Troubleshooting Guide,* or Direction 15491, *Signa Advantage 1.5T & 0.5T System Troubleshooting Guide.* Use the table to record the TLT results of the defective coil.

SITE:	**NAME:**	**DATE:**
TLT FILE NUMBER:		
SNR: NOISE		
SNR MEAN		
SNR SIGNAL		
SNR AREA		
TR MAP: 89–91	%	%
85–95	%	%
65–115	%	%
FLIP MEAN		
FLIP SDV		
RCV MEAN		
RCV SDV		
DIFF MEAN		
DIFF SDV		

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